UTILITY & DEMAND

Objectives

- This chapter will help you to:
  - Describe preferences using the concept of utility and distinguish between total utility and marginal utility
  - Explain the marginal utility theory of consumer choice
  - Use marginal utility theory to predict the effects of changing prices and incomes
  - Explain the paradox of value

Water, Water, Everywhere

- Why is water, which is so vital to life, far cheaper than diamonds, which are not essential?
- Why is the demand for electricity inelastic, while the demand for PCs is elastic?
- This chapter helps you to answer questions such as these.

Household Consumption Choices

- A household's consumption choices are determined by:
  - Consumption possibilities
  - Preferences

- Consumption Possibilities
  - A household's consumption possibilities are constrained (limited) by its budget and the prices of the goods and services it buys.
  - A *budget line* describes the limits to a household's consumption choices.
  - Consider a person's income is $30 spent on soda and/or movies. The price of soda is $3 and the price of movie ticket is $6. Then:

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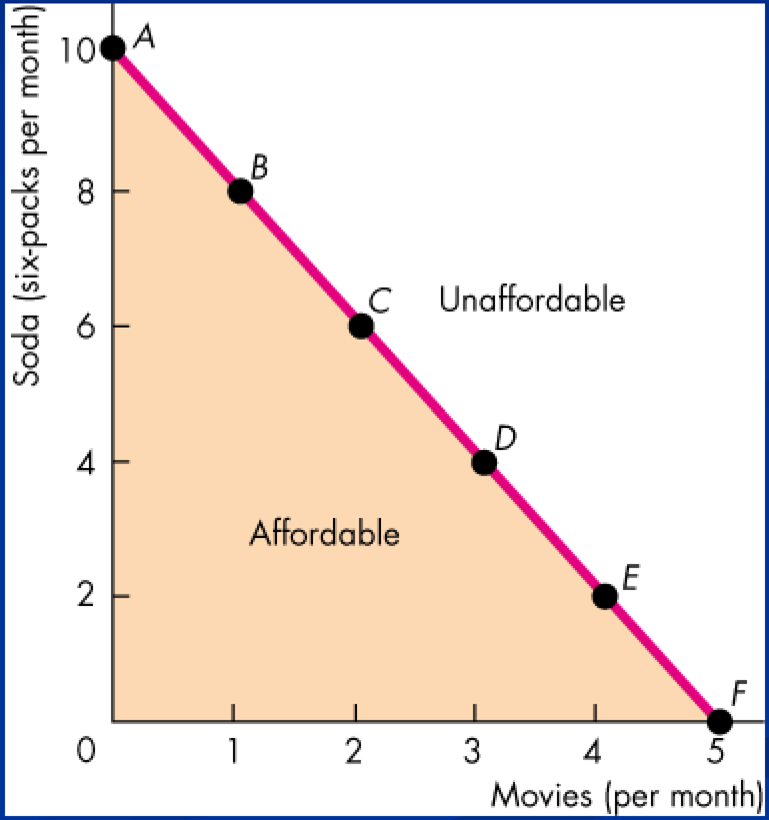
- The household can afford all the points on or below the budget line.
- The household cannot afford the points beyond the budget line.



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- Preferences
  - A household's preferences determine the benefits or satisfaction a person receives from consuming a good or service.
  - The benefit or satisfaction from consuming a good or service is called **utility**.
- Total Utility
  - **Total utility** is the total benefit a person gets from the consumption of goods. Generally, more consumption gives more utility.

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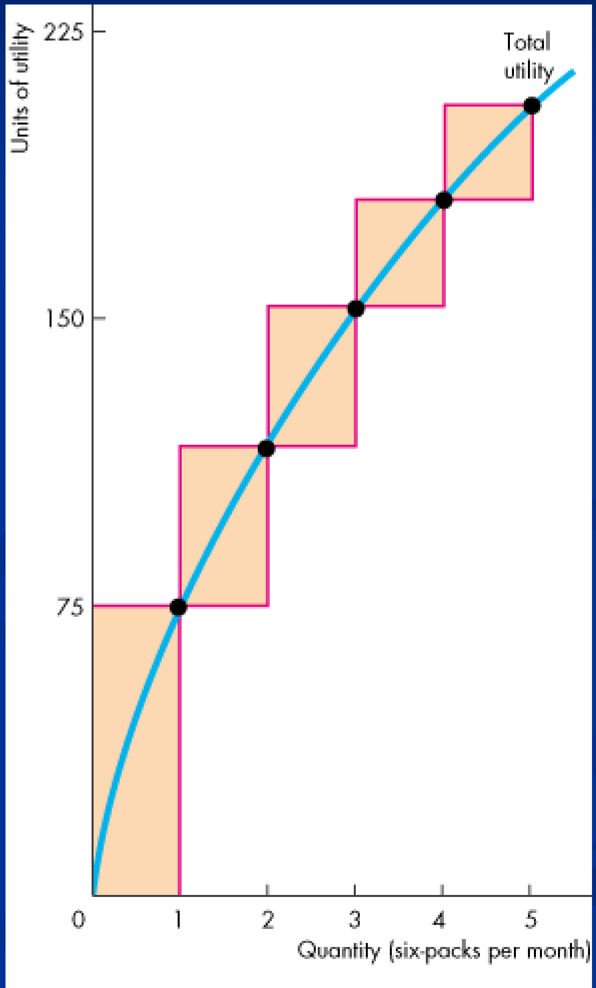
Total utility increases with the consumption of a good.



(a) Total utility

(Cont.)

- Marginal Utility
  - **Marginal utility** is the change in total utility that results from a one-unit increase in the quantity of a good consumed.
  - As the quantity consumed of a good increases, the marginal utility from consuming it decreases.
  - We call this decrease in marginal utility as the quantity of the good consumed increases the principle of **diminishing marginal utility**.

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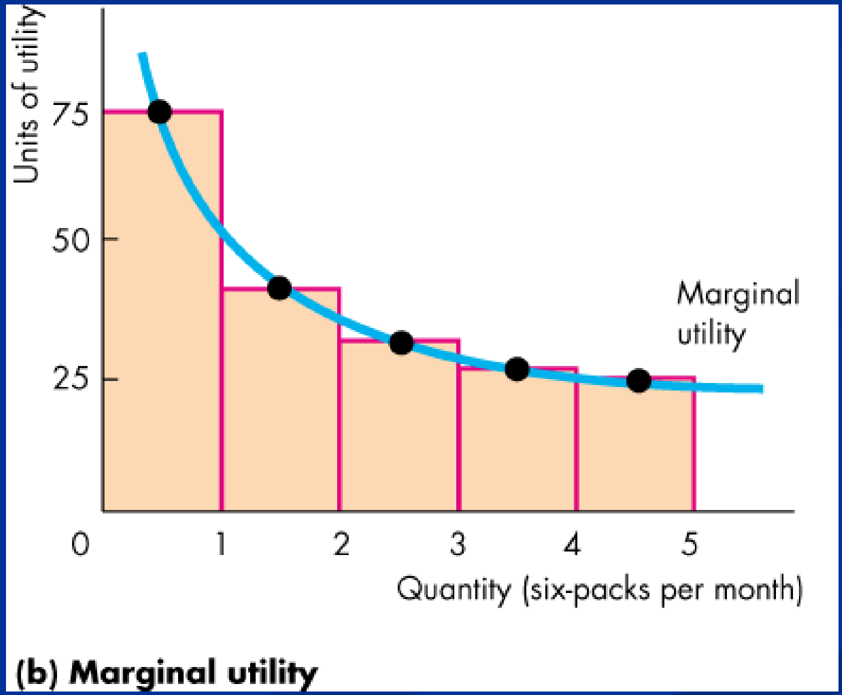
Utility is analogous to temperature.

Both are abstract concepts and both are measured in arbitrary units.



Maximizing Utility:

  - The key assumption of marginal utility theory is that the household chooses the consumption possibility that maximizes total utility.
- The Utility-Maximizing Choice:
  - We can find the utility-maximizing choice by looking at the total utility that arises from each affordable combination.

(Cont.)

- Equalizing Marginal Utility per Dollar Spent
  - Using marginal analysis, a consumer's total utility is maximized by following the rule:
  - Spend all available income and equalize the marginal utility per dollar spent on all goods.
  - The **marginal utility per dollar spent** is the marginal utility from a good divided by its price.

(Cont.)

- Call the marginal utility of movies MU_M
- Call the marginal utility of soda MU_S
- Call the price of movies P_M
- Call the price of soda P_S
- The marginal utility per dollar spent on movies is MU_M/P_M
- The marginal utility per dollar spent on soda is MU_S/P_S

(Cont.)

- Total utility is maximized when:
  - $MU_M/P_M = MU_S/P_S$

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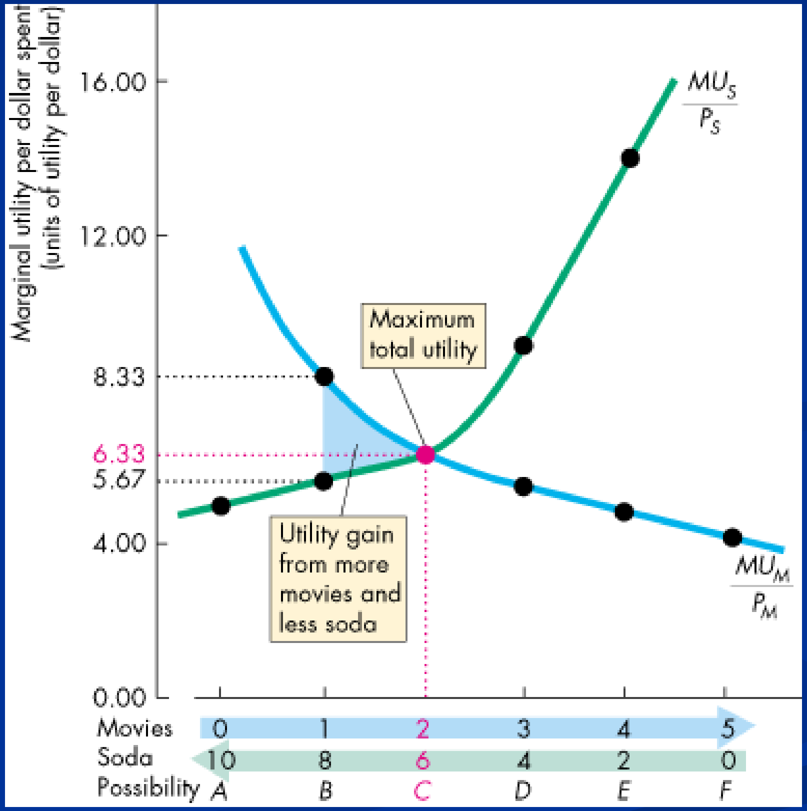
- If $MU_M/P_M > MU_S/P_S$, then moving a dollar from soda to movies increases the total utility from movies by more than it decreases the total utility from soda, so total utility increases.



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Only when $MU_M/P_M = MU_S/P_S$, is it not possible to reallocate the budget and increase total utility.

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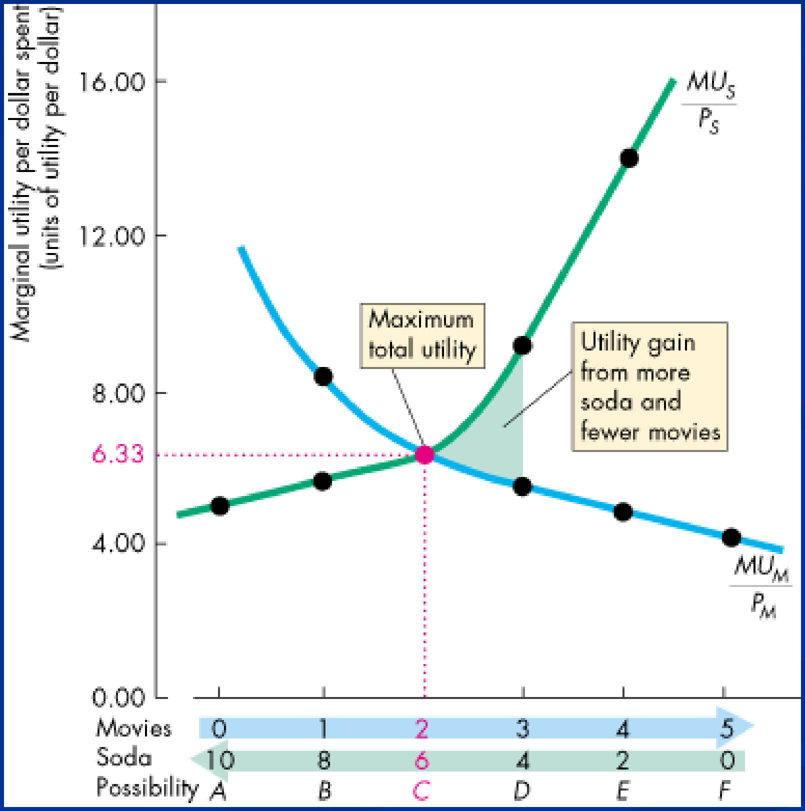
- Similarly, if $MU_S/P_S > MU_M/P_M$, then moving a dollar from movies to soda increases the total utility from soda by more than it decreases the total utility from movies, so total utility increases.



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Again, only when $MU_M/P_M = MU_S/P_S$, is it not possible to reallocate the budget and increase total utility.

Numerical Illustration:

Assume price of X = 2; Price of Y = 10

Q_X	TU_X	MU_X	$\frac{MU_X}{P_X}$	Q_Y	TU_Y	MU_Y	$\frac{MU_Y}{P_Y}$
1	30	30	15	1	50	50	5
2	39	9	4.5	2	95	45	4.5
3	45	6	3	3	148	43	4.3
4	50	5	2.5	4	178	30	3
5	54	4	2	5	198	20	2
6	56	2	1	6	213	15	1.5

TU is different for the different combinations. Choice of which combination will depend on income available.

Predictions of Marginal Utility Theory:

- A Fall in the Price of a Movie
  - When the price of a good falls the quantity demanded of that good increases—the demand curve slopes downward.
  - For example, if the price of a movie falls, we know that MU_M/P_M rises, so before the consumer changes the quantities consumed, $MU_M/P_M > MU_S/P_S$.
  - To restore consumer equilibrium (maximum total utility), the consumer increases the quantity of movies consumed to drive down the MU_M and restore $MU_M/P_M = MU_S/P_S$.

(Cont.)

- A change in the price of one good changes the demand for another good.
- You've seen that if the price of a movie falls, MU_M/P_M rises, so before the consumer changes the quantities consumed, $MU_M/P_M > MU_S/P_S$.
- To restore consumer equilibrium (maximum total utility) the consumer decreases the quantity of soda consumed to drive up the MU_S and restore $MU_M/P_M = MU_S/P_S$.

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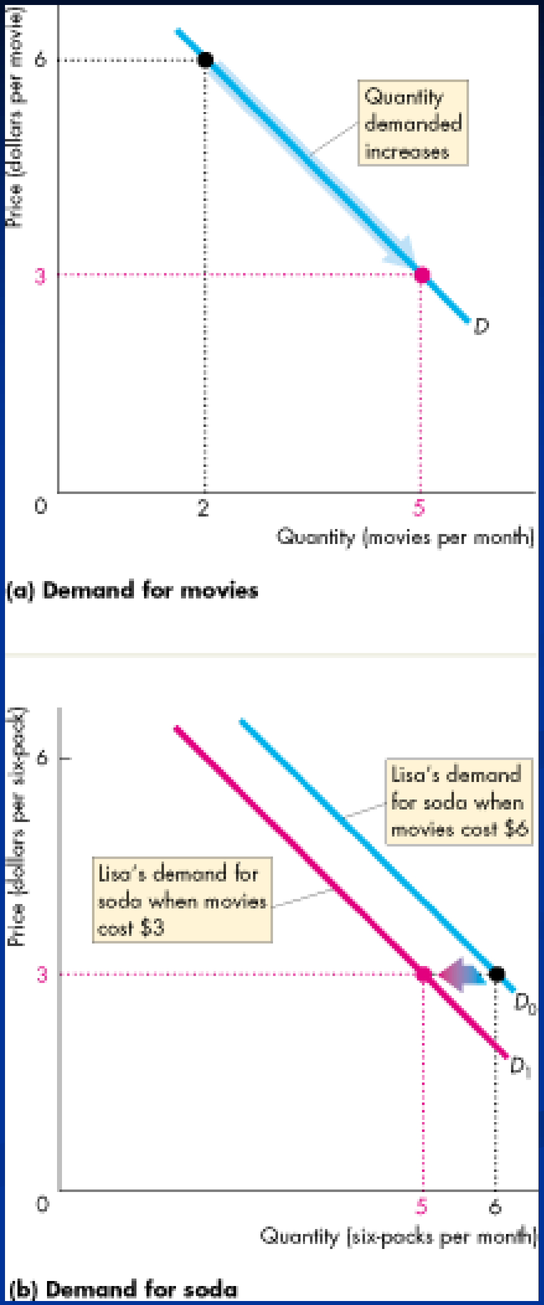
A fall in the price of a movie increases the quantity of movies demanded—a movement along the demand curve for movies, and decreases the demand for soda—a shift of the demand curve for soda.



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- A Rise in the Price of Soda
  - Now suppose the price of soda rises.
  - We know that MU_S/P_S falls, so before the consumer changes the quantities consumed, $MU_S/P_S <$ MU_M/P_M.
  - To restore consumer equilibrium (maximum total utility) the consumer decreases the quantity of soda consumed to drive up the MU_S and increases the quantity of movies consumed to drive down $MU_{M.}$ These changes restore $MU_M/P_M = MU_S/P_S$.

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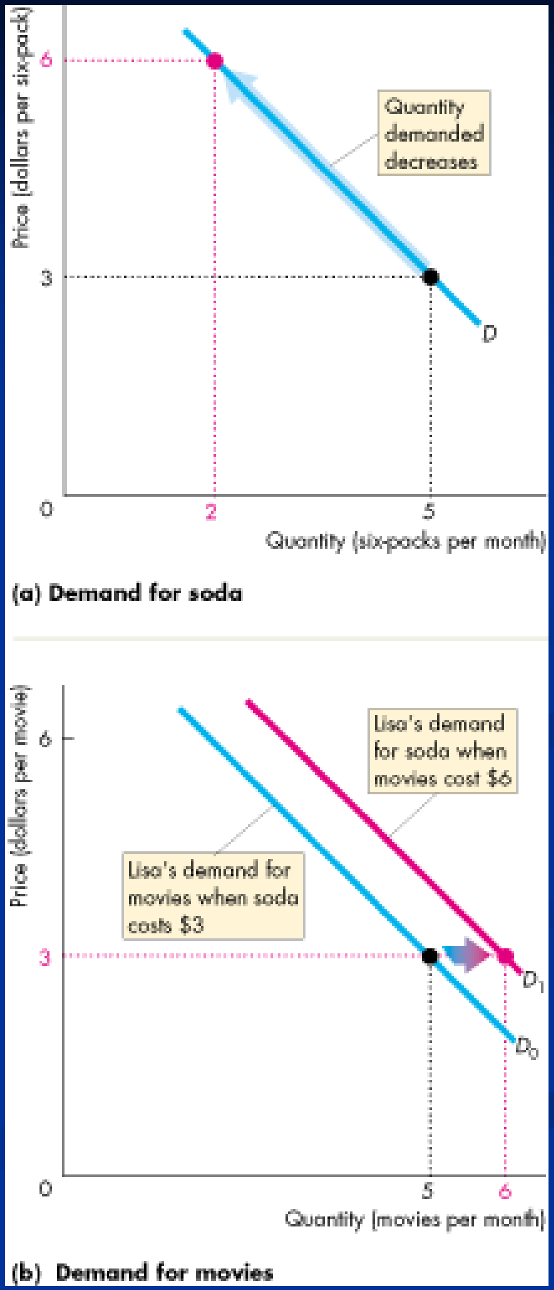
A rise in the price of soda decreases the quantity of soda demanded—a movement along the demand curve for soda, and increases the demand for movies—a shift of the demand curve for movies.



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- Individual Demand and Market Demand
  - The market demand for a good is the relationship between the price of the good and total quantity demanded of that good.
  - The individual demand for a good is the relationship between the price of the good and the quantity demanded by one person.

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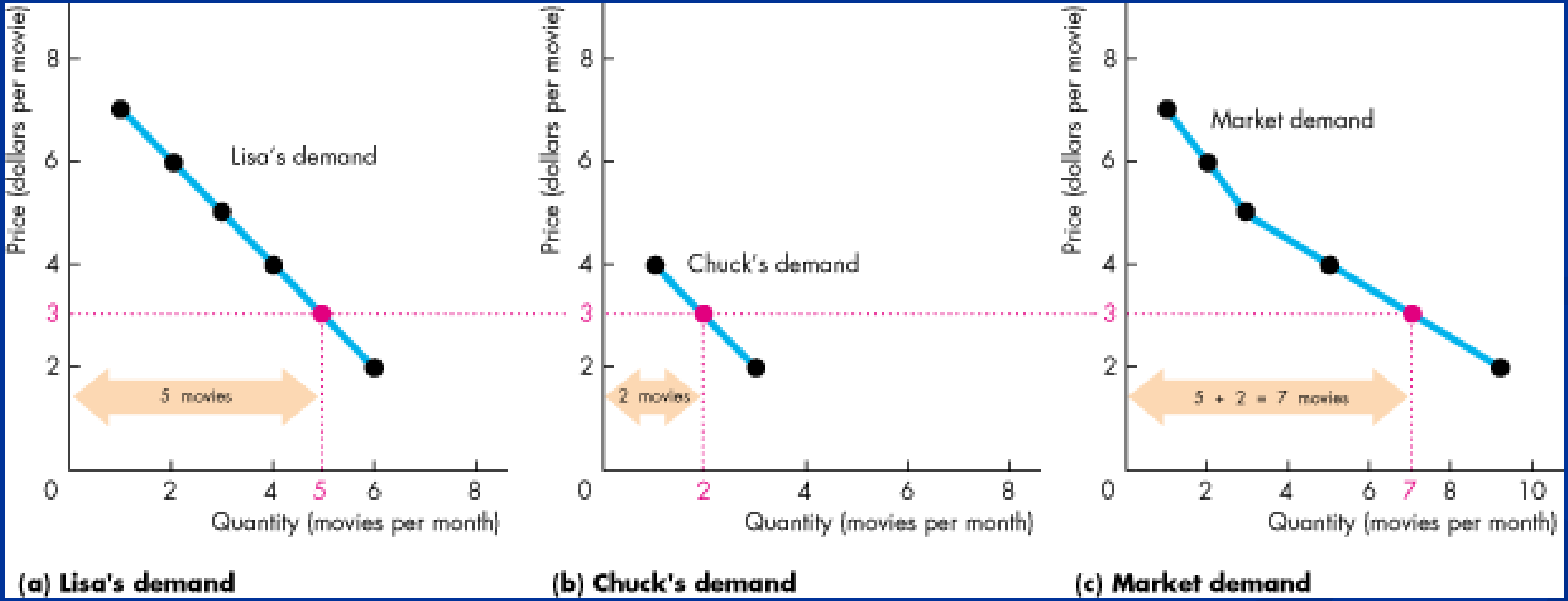
Price (dollars per movie)
Lisa's demand
5 movies
Quantity (movies per month)
Chuck's demand
2 movies
Market demand
5 + 2 = 7 movies
(a) Lisa's demand
(b) Chuck's demand
(c) Market demand

(Cont.)

- Marginal Utility and Elasticity
  - We can predict the price elasticity of demand for a good by knowing the characteristics of the marginal utility of the good.
  - If as the quantity consumed, marginal utility diminishes rapidly, then a given price change will bring a small quantity change to restore consumer equilibrium, and demand will be inelastic.

The Paradox of Value:

- The paradox of value "Why is water, which is essential to life, far cheaper than diamonds, which are not essential?" is resolved by distinguishing between total utility and marginal utility.

(Cont.)

- Because water is plentiful and we consume much of it, total utility is large but the marginal utility of an extra unit (liter) is small. Hence, according to the law of diminishing marginal utility, consumers will be willing to pay only a small price for the extra (marginal) unit.
- If consumers are offered an all-or-nothing choice for water consumption (e.g. SDG100 for 1000 gallons or no water) they will conserve a lot on water consumption. Hence, the first few gallons will have a high marginal utility and therefore a high price. This is the reason why water yields a large consumer surplus.
- Diamonds, on the other hand, are scarce, and according to the law of diminishing marginal utility their marginal utility will be high (total utility small) and hence consumers will be willing to pay a high price for them.

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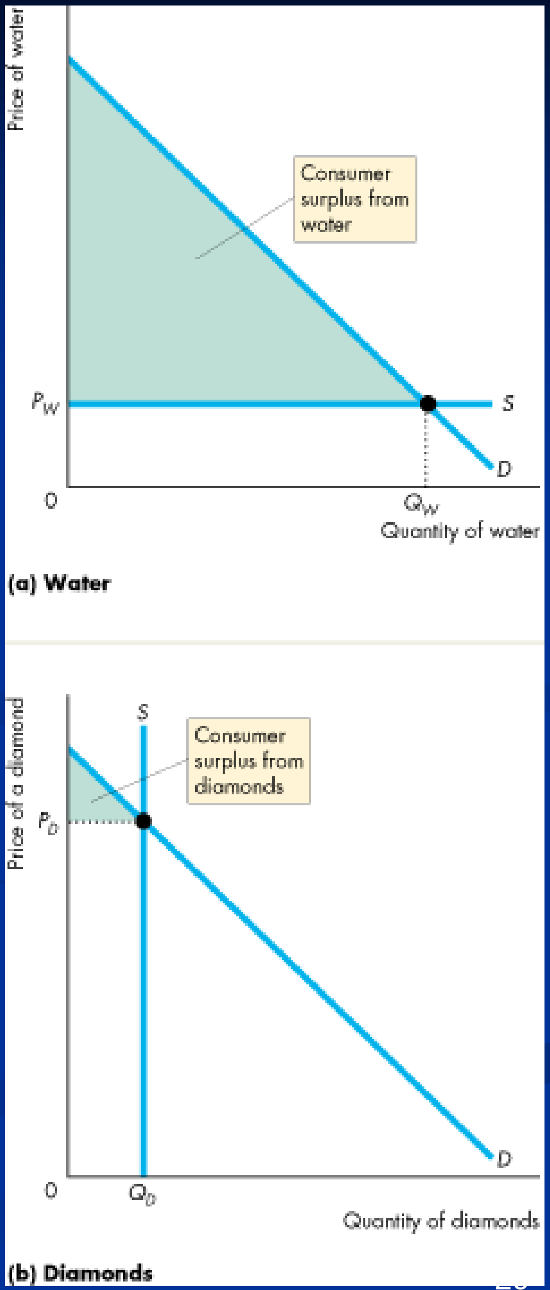
- The total utility and consumer surplus from water is large but the marginal utility and price of water is small.

In contrast, the total utility and consumer surplus from diamonds is small but the marginal utility and price of a diamond is large.



Readings: Samuelson & Nordhaus, Ch. 5